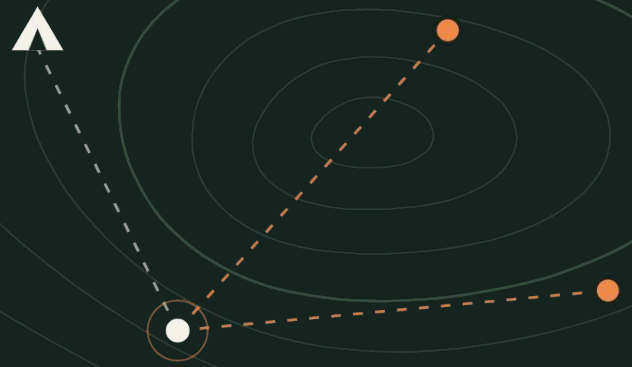
SDP27 · PRELIMINARY DESIGN REVIEW

LODESTONE

Smart Compass



TEAM #16

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Problem Statement

Cell service is often unavailable in remote hiking and camping areas, leaving separated group members unable to see each other's locations and without a last known position if someone fails to return.

Our goal is to create a low-cost, long-battery-life smart compass watch that points you directly to the people you're with, off-grid, using efficient LoRa radio and a simple glanceable display.



Project Goals

G1

Off-Grid Group Localization

Implement peer-to-peer position sharing between wearable nodes over a long range link, with no reliance on cellular or internet connectivity

G2

Waypoint Storage:

Allow the user to save a GNSS waypoint (e.g., a campsite) in memory with output that is visible on the directional display.

G3

Low-Power Operation and Battery Life

Achieve a full day of outdoor use on a single charge through an on-demand display wake and GNSS and LoRa transmission

G4

Wearable Form Factor and Display Usability:

Design a compact, lightweight wrist enclosure with a display that conveys direction through a single glanceable indicator

Preliminary Design



Survey of Existing Solutions

Existing Solution #1: Garmin Tactix 8 AMOLED

- Up to 29 days of battery in smartwatch mode
- Military standard durability
- Topoactive terrain maps, customizable map layers, and course navigation
- Track other users using GroupTrack and LiveTrack



Limitations:

- High cost, $1399.99 + 69.99/year
- Outdoor Maps+ requires an annual subscription
- Tracking caveat
- Requires users to have their smartphone on them



Survey of Existing Solutions

Existing Solution #2: Totem Compass

- Wearable that points to friends with colored lights
- No phone, Wifi, or cell service required
- GPS used for positioning, and 1 km range*
- About 10 hours of battery life in compass mode



Limitations:

- Around $120 each unit
- No display, map, or messaging; Direction lights only
- Short battery life
- Limited use, range
- Main use is for concerts

* Vendor claim in open terrain

Survey of Existing Solutions

Existing Solution #3: Lynq

- Digital compass that points towards friends
- No cell or WiFi needed
- 8.8 km range*
- Small round monochrome screen
- About 24 hours of active use



Limitations:

- Very minimal display, only shows general direction and distance of one target
- GPS signal issues, especially in densely wooded areas
- Up to 5 min wait times when initializing two Lynq devices

* Vendor claim in open terrain

Side-by-side Comparison

Criterion	Garmin Tactix 8 AMOLED	Totem Compass	Lynq	Our target
Works with no cell service	Yes	Yes	Yes	Yes
Shows friends multiple friends locations	Yes w/ subscription	Yes	One Friend	Yes
Guides back to a saved spot	Yes	No	No	Yes
Worn on the wrist	Yes	No	No	Yes
Range between users	40 km*	1 km*	8.8 km*	≥ 0.5 km, stretch 1km
Battery life	29 days	Up to 10 h	24 h	≥ 24 h active use
Cost	$1,399.99 + 69.99 annual	$120	$124	≤ $120
Weight	96 g	54 g	74 g	≤ 100 g

Takeaway: No existing option combines reliable off-grid communication, a wrist-worn form factor, and low cost; our project will fill that gap.

* Vendor claim in open terrain

Design Specifications - Navigation

S1: Range

- Stable connection at 500 m, stretch 1 km. Why: sized to a group splitting at a junction or campsite; competitors claim 1 - 4.8 km

S2: Link-up time

- ≤ 60 s after both units have a GNSS fix until the first friend is shown. Why: Lynq users report waits up to 5 min, 60 seconds is much better

S3: Heading

- Displayed direction within ±10 degrees of true bearing. Why: sensor error of about 2–4.5 degrees leaves margin for mounting and calibration

S4: Position

- Horizontal error ≤10 m over 10 minutes in open air. Why: this is the typical consumer GNSS in open sky

Design Specifications - Hardware

S5: Battery Life

- ≥ 24 hours per charge in use. Why: some competitors also claim about 24 hrs of use.

S6: Group Size

- ≥ 3 users. Why: enough for a small hiking party.

S7: Cost

- Projected BOM ≤ $120 per unit. Why: should beat cheapest competitor's $120

S8: Size and Weight

- ≤ 100 g with strap and case; case no larger than 50 × 50 × 20 mm. Why: normal watch form factor

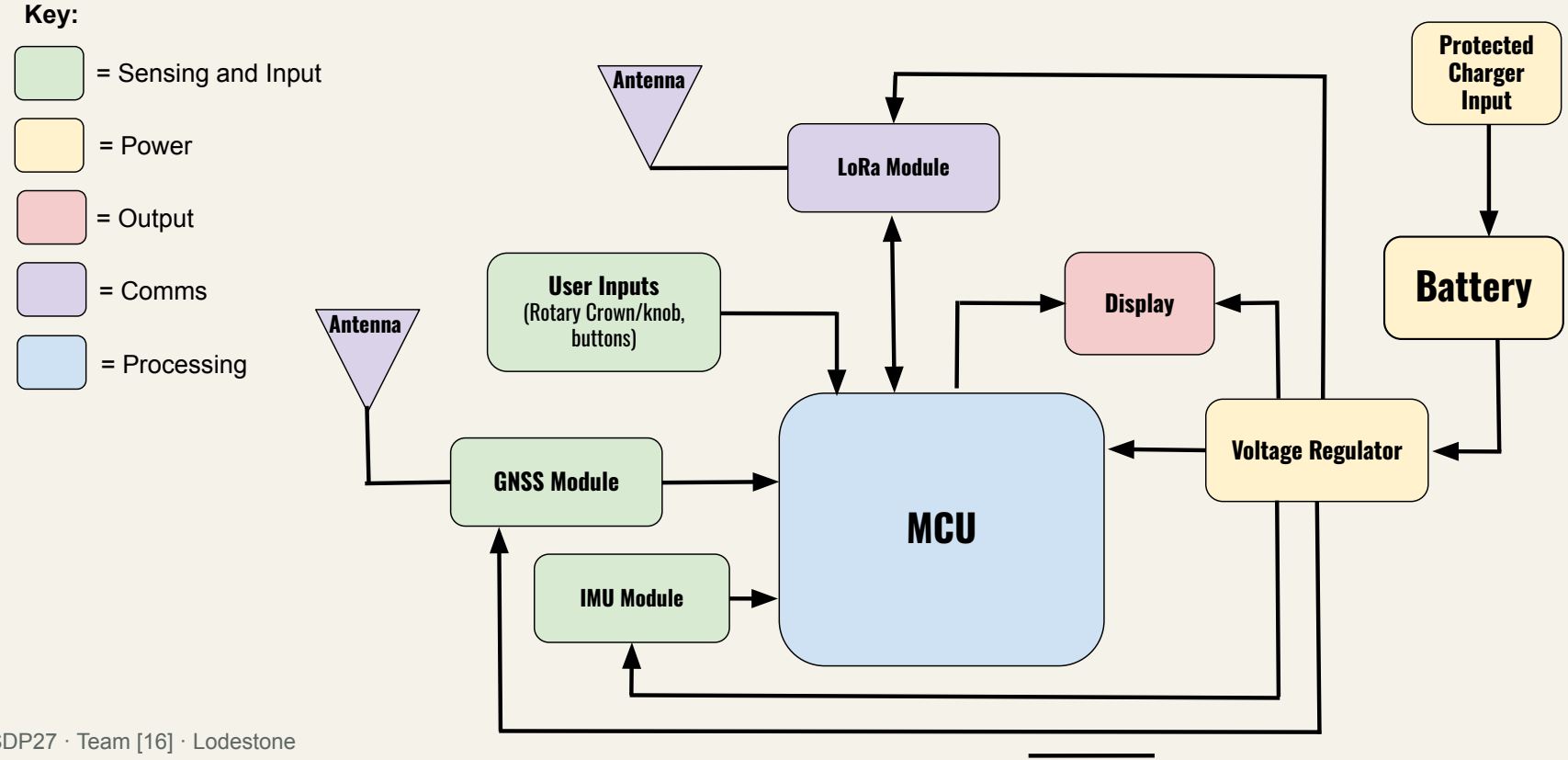
S9: Safety

- Passes tests: reverse polarity, over-voltage/over-current. Why: LiPo specifications for course to ensure safety during operation.

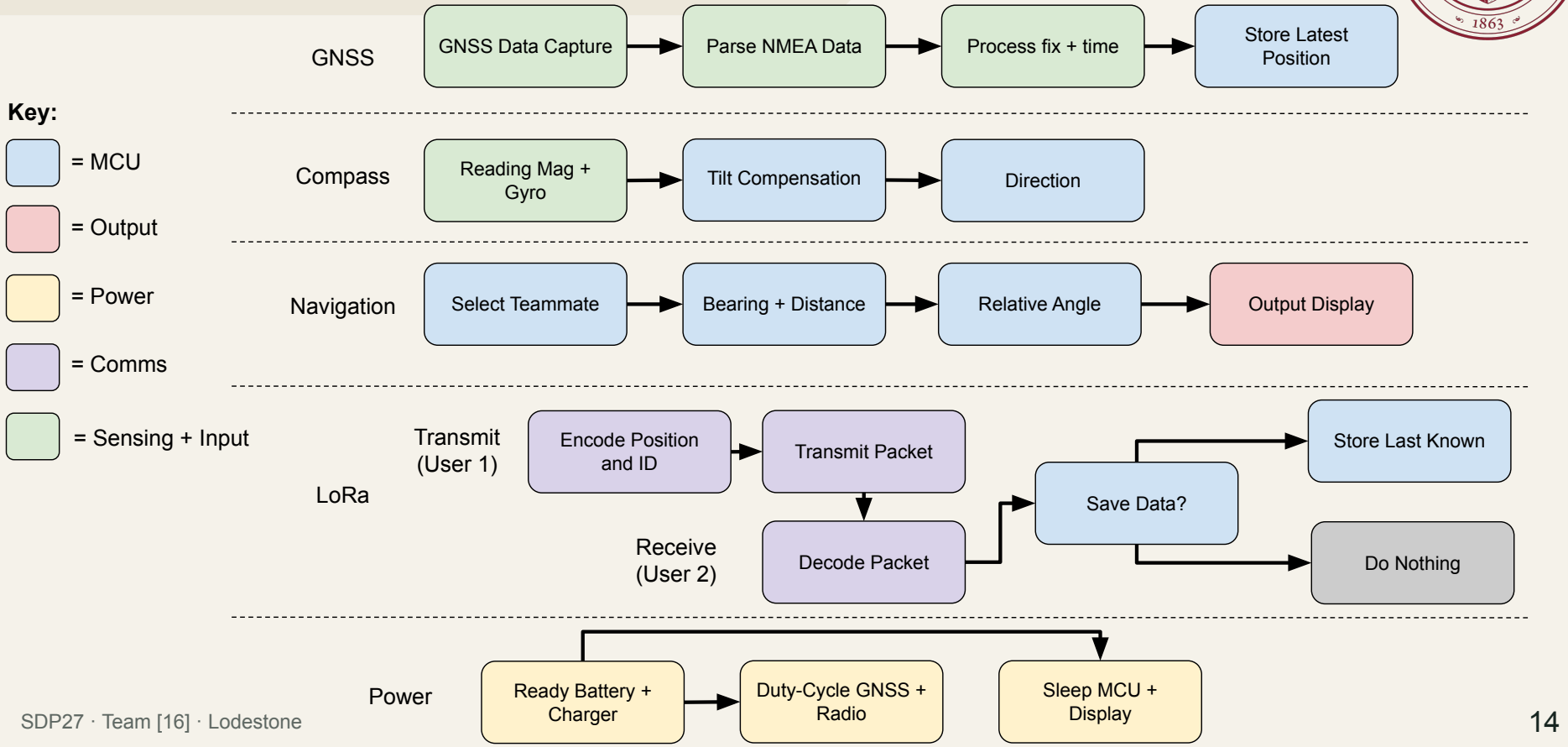
Test Plan

Spec	Procedure	Pass	Spec	Procedure	Pass
S1: Range	Fixed beacon sends numbered packets; the compass logs RSSI/SNR at 100, 250, 500, 1000 m; compared to phone GPS	≥90% of packets delivered at 500 m	S6: Group Size	Show 2 users positions on a single device to show functionality of multiple users	2 users shown
S2: Link-up Time	Power on both units outdoors; timer from both GNSS fixes to the first dot appearing	Median ≤60 s	S7: Cost	Sum the 1,000-unit distributor prices	≤$120
S3: Heading	Hold the unit level at 5 headings, with 3 trials each. Compare it with phone compass	≥13/15 tests within ±10 degrees	S8: Size, Weight	Check that CAD/KiCad models stay within boundaries. Measure with scale and calipers	≤100 g; ≤50×50×20 mm
S4: Position:	Static log for 10 min at a phone-referenced point; compute the 95th percentile offline	95% within 10 m	S9: Safety	Test overvoltage and overcurrent with bench supply, reverse the polarity	No damage to device or battery
S5: Battery Life	MDR: Measure a test unit on battery from full charge to dead.	≥24 h			

Hardware Block Diagram



Software Block Diagram



Design Decisions

Part/Technology	Decision	Reasoning
Radio	LoRa 915 MHz	We chose Lora 915 MHz over alternatives like BLE, GMRS/FRS, or satellite. BLE does not have the range, GMRS/FRS is too expensive and power demanding, and satellite requires a subscription to use
MCU	nRF54 vs. nRF52	nRF series Raytac modules have BLE, SPI and I2C, and excellent power savings in a very small module. nRF52 has more extensive libraries but higher power draw
IMU	BNO085	Module has on-chip fusion and calibration, about 2 deg error well below 10 deg goal
GNSS	u-blox MAX-M10S module	Module has <25 mW tracking, works with a passive antenna, has a Zephyr compatibility
LoRa Module	SX1262 @915 MHz	In-tree Zephyr driver, lower receive current than the SX1276, +22 dBm output.
Display	MIP LED	Memory-in-pixel panel: reflective, readable in sun, very low power.
Power Supply	Lithium Ion Battery + USB-C charger	BQ25185 charger, protected 500 mAh LiPo, one 3.3 V rail for peripherals and MCU

MDR Deliverables and PCB Plans

Our prototype aims to demonstrate that the three core functions work on real hardware: knowing which way you're facing, knowing where you are, and sharing that position with teammates over a long-range radio link, with no cell service.

MDR Prototype:

- One compass node on MCU with IMU, GNSS Module, LoRa Module, and a display, plus one fixed beacon node (MCU + LoRa). Simulated a teammate connection.
- Displays live heading at different orientations, logs outdoor GNSS error, logs range and delivery rate at 100-1000 m, shows ability to save + pathfind to waypoints, meets 24 hour battery life projection

PCB Plans:

- Schematic is due Nov 6 and layout Nov 13, ordered about Nov 16.
- PCB v1 is brought up after MDR.

Bill of Materials

MDR Prototypes (1x full compass, 1x dummy beacon)

Part Name	Possible Choice	Cost
MCU	nrf54L15-DK	$43
IMU	BNO085	$13
LoRa, LoRa MCU	Wio-SX1262, XIO nrf54L15	2 x $21
GNSS	MAX-M10S breakout	$46
GNSS Antenna	Sparkfun GNSS SMA Antenna	$16
Battery	Adafruit LiPo (500mAH)	$8
Display	GC9A01 240×240px display 1.28”	$10
Power Supply	Lithium Ion Battery	$7
Total	Course budget: $ 500	$185

Final compass estimate, per unit

Part Name	Notes	Cost
MCU	Raytac nRF54L15 module	$5
LoRa module	SX1262 + strap antenna	$5
Display	MIP LED + front light	$35
GNSS	MAX-M10S and chip antenna	$17
GNSS antenna	Johanson 1575AT44A0010	$2
IMU	BNO085	$13
Power Supply	Adafruit LiPo (500mAH)	$7
Power Management	BQ25185 Charger	$5
LoRa Antenna	Strap FPC antenna	$10
Other	Passives, PCB + Stencil	$20
Total	Course budget: $ 500	$119

Gantt Chart

Task	Lead (s)	10/06-10/12	10/13-10/19	10/20-10/26	10/27-11/02	11/10-11/16	11/17-11/23	11/24-11/30	12/01-12/07
Setup, Component selection, prototype parts ordering	Owen + Wil								
Display and UI implementation	Quinn + Eric								
PCB schematic, layout, and parts ordering	Owen + Quinn								
Firmware development: GNSS, LoRa, IMU integration	Jack + Eric + Wil								
Enclosure CAD and prototyping	Jack + Owen								
System integration and validation testing	Eric + Wil								

Oct 27: PDR

Nov 1: Schematic Due

Nov 13: Layout Due

Nov 30 - Dec 4: MDR

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Questions?

